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(US Green-Card holder)

Mohammad Javadi Balakan

Location: San Jose, CA

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Employment History

Research Scientist (2025 - Present)

Postdoctoral Fellow (2022 - 2024)

College of Nanotechnology, Science, and Engineering (CNSE)

State University of New York, University at Albany

Topic: Topological Superconductors for Fault-Tolerant Quantum Computation

Postdoctoral Associate (2019 - 2021)

Department of Physics, University of Tehran, Iran

Topic: CMOS-compatible Solid-State Devices Based on 2D Materials

Visiting Researcher (Mar. 2018 - Sep. 2018)

Department of Physics, Sapienza Università di Roma, Italy

Supervisor: Prof. Giorgio Parisi

Topic: Statistical properties of harmonic spheres near jamming

Education

PhD in Solid-State Physics, University of Tehran, Iran (2019)

Thesis: Charge Transport in Metal-Oxide Nano-Structures for Photovoltaic and Optoelectronic Applications

MSc in Nanoscience and Nanotechnology, University of Tehran, Iran (2014)

Thesis: Electron Transport in Nano-Structured Solar Cells

BSc in Physics, Urmia University, Iran (2012)

Awards and Honors

- Chamran Postdoctoral Fellowship, Iran National Elites Foundation (INEF), 2020
- Outstanding PhD Thesis in Physics (nationwide), Physical Society of Iran, 2019
- Outstanding PhD Dissertation in Physical Sciences, University of Tehran, 2019
- INEF Graduate Scholarship, 2016 - 2018
- Ranked 10th nationally in the PhD entrance examination, 2014
- Graduated with distinction, MSc in Physics, 2014
- Ranked 19th nationally in the MSc entrance examination, 2012
- Graduated with distinction, BSc in Physics, 2012

Technical Skills

Device fabrication and structural characterization: cleanroom nanofabrication, process development, electron-beam lithography (EBL), focused ion beam (FIB), scanning electron microscopy (SEM), atomic force microscopy (AFM), reactive ion etching (RIE), Raman and optical spectroscopy, thin film deposition (PVD, e-beam evaporation, CVD, PECVD), 2D materials, wide bandgap oxides, organic semiconductors, topological insulators, and topological superconductors.

Electrical characterization: DC/AC semiconductor parameter analysis (SMU, LCR, etc.), optoelectronic characterization tests (photovoltaic cells, optical detectors/sensors), low-noise/precision measurements (lock-in amplifiers, Atto-sense modules, low-temperature LPFs), cryogenic systems, quantum transport, phase-sensitive magnetic/flux measurements.

Computation, data analysis, and automation: Python, Mathematica, LabVIEW, KLayout (layout/mask design), statistical analysis, instrument automation.

Professional Service

- Peer review for scientific journals including Physical Review Letters, 2D Materials, Applied Physics Letters, and Nanotechnology (ORCID).
- Served as a committee member for the Physical Society of Iran symposium, 2018.

Research Experience

CNSE, University at Albany, USA (Prof. Ji Ung Lee's lab)
Topological superconductivity for fault-tolerant quantum computation

- Electrical transport studies of topological superconductors in collaboration with Stony Brook University and BNL (Prof. Qiang Li and Prof. Genda Gu).
- Developed processes for end-to-end fabrication of nanoscale quantum devices.
- Upgraded a closed-cycle cryostat, designed and implemented cryogenic low-pass filters (LPFs) for precision electrical measurements, and developed Python scripts for instrument automation.
- Demonstrated half-integer flux quantization, intrinsic magnetization of the superconducting condensate, and Aharonov-Bohm oscillations of edge states in mesoscopic Fe(Te,Se) devices.

Laboratory for nanotechnology, University of Tehran, Iran (Prof. Abdi's lab)
CMOS-compatible solid-state devices based on 2D materials

- Investigated charge transport across graphene-silicon 2D/3D Schottky junctions.
- Developed a model for the fundamental efficiency of graphene-based solar cells.
- Fabricated and characterized infrared detectors based on 2D MoS2 layers.
- Developed scalable position-sensitive detectors based on hybrid organic/inorganic PEDOT:PSS/Silicon heterojunctions (*APL Editor's Pick*).
- Pioneered a scalable method for synthesis of 2D borophene metallic layers using chemical vapor deposition.

Department of Physics, University of Tehran (Prof. Abdi's and Prof. Arzi's joint group)

- Fabrication and characterization of nanostructured solar cells based on dye-sensitized solar cells (solar simulator, IV, quantum efficiency, carrier lifetime and diffusion length measurement).
- Monte-Carlo simulation of charge transport in disordered wide band-gap semiconductors for photovoltaic and optoelectronic applications.
- Computer modeling for PECVD growth of single-standing carbon nanotubes.

Nanoelectronic laboratory, Tarbiat Modares University, Iran (Dr. Sara Darbari's lab)

- Investigated sound-driven CNT-based tribo-electric nanogenerator in sonic array.
- Developed a piezo-phototronic device based on graphene/MoS2 heterostructure.

Teaching And Mentoring

Teaching Assistant, University of Tehran (2012–2018): Quantum Mechanics (undergraduate), Advanced Solid-State Physics & Laboratory (graduate), Advanced Quantum Mechanics (graduate)

- Mentored Ph.D. researcher Daniel Titcomb in experimental studies of topological superconductors (Dr. Lee's group).
- Mentored graduate researcher AliAkbar Noroozi in graphene-silicon optical detectors; work contributed to a Ph.D. thesis and two publications (Dr. Abdi's group).
- Mentored graduate student Mahdieh Gholami on CVD growth of two-dimensional MoS_2 for infrared detectors; work contributed to a master's thesis and a publication (Dr. Abdi's group).

Invited Talks

- *"Half-integer winding of order parameter & intrinsic magnetization of superconducting condensate in Fe(Te,Se)"*, GRC correlated electron systems, Mount Holyoke College, MA, USA (2026).
- "*Topological superconductivity in Fe(Te,Se)*", Laboratory for Physical Sciences (LPS), College Park, MD, USA (2025).
- "*Charge transport at graphene-silicon 2D/3D interface*", *TMU*, Tehran, Iran (2020).
- "*Negative photoconductivity in ZnO nanowires*", IPM, Tehran, Iran (2018).

Publications

- M. J. Balakan, S. Heidari, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and J. U. Lee, Intrinsic magnetization of the superconducting condensate in Fe(Te,Se), will be published in Phys. Rev. B. Letters (2026) [**Editors' suggestion**].
- M. J. Balakan, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and J. U. Lee, Half-integer winding of order parameter in an iron-based superconductor, submitted (2026).
- M. J. Balakan, S. Heidari, G. Gu, Q. Li, D. Titcombe, Z. Xiao, H. Frost, E. Bhatia, C. Kow, P. Rao, K. Watanabe, T. Taniguchi, and J. U. Lee, Aharonov–Bohm interference from coherent spin-polarized edge transport in Fe(Te,Se) superconducting rings, submitted (2025).
- M. J. Balakan, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and J. U. Lee, Observation of half-quantum vorticity in an iron-based superconductor, (2025) [arXiv:2505.01522].
- A. Mazaheri, M. Javadi, and Y. Abdi, Chemical vapor deposition of two-dimensional boron sheets by thermal decomposition of diborane, *ACS Applied Materials & Interfaces* 13(7), 8844-8850, (2021).

- M. Javadi, AA. Noroozi, and Y. Abdi, Kinetics of charge carriers across a graphene–silicon Schottky junction, *Physical Review Applied* 14, 064048, (2020).
- AH. Mosaddegh, AA. Noroozi, M. Javadi, and Y. Abdi, Ultrahigh photovoltage responsivity of PEDOT:PSS–silicon hybrid heterojunction photodiodes, *Applied Physics Letters* 117, 073301, (2020).
- M. Javadi, Theoretical efficiency limit of graphene–semiconductor solar cells, *Applied Physics Letters* 117, 053902, (2020).
- M. Javadi, A. Mazaheri, H. Torbatiyan, Y. Abdi, Mechanism of charge transport in hybrid organic–inorganic PEDOT:PSS/silicon heterojunctions, *Physical Review Applied* 12, 034002, (2019).
- M. Javadi, AA. Noroozi, A. Mazaheri, Y. Abdi, Sequentially assembled graphene layers on silicon, the role of uncertainty principles in graphene–silicon Schottky junctions, *Advanced Optical Materials* 7, 1900470, (2019).
- M. Javadi, Y. Abdi, Frequency-driven bulk-to-surface transition of conductivity in ZnO nanowires, *Applied Physics Letters* 113, 051603, (2018).
- M. Javadi, M. Gholami, H. Torbatiyan, Y. Abdi, Hybrid organic/inorganic position-sensitive detectors based on PEDOT:PSS/n-Si, *Applied Physics Letters* 112, 113302, (2018) [**Editors' Pick**].
- M. Javadi, M. Gholami, Y. Abdi, IR position-sensitive detectors based on double-junction asymmetric TiO_2/MoS_2/reduced graphene-oxide sandwiches, *Journal of Material Chemistry C* 6, 8444, (2018).
- M. Javadi, A. Heidari, S. Darbari, Realization of enhanced sound-driven CNT-based triboelectric nanogenerator, utilizing sonic array configuration, *Current Applied Physics* 18, 361, (2018).
- M. Javadi, H. Torbatiyan, Y. Abdi, Ultra-low intensity UV detection using partitioned mesoporous TiO_2, *Applied Physics Letters* 111, 063113, (2017).
- M. Javadi, Y. Abdi, Kinetic Monte Carlo simulation of single-electron multiple-trapping transport in disordered media, *Computer Physics Communications* 221, 282, (2017).
- M. Javadi, Y. Abdi, Nanopattern formation using localized plasma for growth of single-standing carbon nanotubes, *Journal of Nanoparticle Research* 19, 17, (2017).
- M. Javadi, S. Alizadeh, Y. Khosravi, Y. Abdi, Electron transport in Q2D porous network of titania nanoparticles, incorporating electrical and optical advantages in dye-sensitized solar cells, *ChemPhysChem* 17, 3542, (2016).
- M. Javadi, S. Darbari, Y. Abdi, F. Ghasemi, Realization of a piezo-phototronic device based on reduced graphene-oxide/MoS_2 heterostructure, *IEEE Electron Device Letters* 37, 677, (2016).
- M. Javadi, Y. Abdi, E. Arzi, Local collection efficiency in nano-crystalline solar cells, *Solar Energy* 133, 549, (2016).
- M. Sajedi Alvar, M. Javadi, Y. Abdi, E. Arzi, Enhancing the electron lifetime and diffusion coefficient in dye-sensitized solar cells by patterning the layer of titanium dioxide nanoparticles, *Journal of Applied Physics* 119, 114302, (2016).

- M. Javadi, and Y. Abdi, Monte Carlo random walk simulation of electron transport in confined porous TiO_2 as a promising candidate for photo-electrode of nano-crystalline solar cells, *Journal of Applied Physics* 118, 064304, (2015).

Conference Presentations

- M. J. Balakan, Half-integer winding of the order parameter and intrinsic magnetization of the superconducting condensate in mesoscopic Fe(Te,Se) devices, GRC Correlated Electron Systems (2026).
- M. J. Balakan, G. Gu, Q. Li, K. Watanabe, T. Taniguchi, and JU. Lee, Topological superconductivity in Fe(Te,Se) mesoscopic devices, APS Global Meeting (2026).
- Additional presentations at Iranian Physical Society meetings and nanostructured solar cell conferences (2014–2018).